Scaling laws: Test time compute

At first, it was thought that the way to achieve a high Performance Metric was via using models with a large number of parameters.

The Scaling Law showed this to not be true

- defines the Performance as a function of number of parameters and number of tokens consumed during training
- thus, a small and large model might achieve the same Performance
    - with the smaller model needing to consume many more tokens during training

But there is another dimension to consider in order to reach a Performance target

- amount of compute used at *test/inference* time.

Using test time compute to improve Performance

An LLM can produce more than one solution

When predicting the next token (using a Transformer in auto-regressive mode) the Language Model outputs

$$p(\mathbf{y}_{(t)} \mid \mathbf{y}_{(1:t-1)})$$

- a *probability distribution*
- from which we sample a single token $\mathbf{y}_{(t)}$

If we sample deterministically (e.g., $\mathbf{y}_{(t)} = \mathbf{y}_{j^*}$ where $j^* = \underset{t}{\operatorname{argmax}}\, p(\mathbf{y}_{(t)} \mid \mathbf{y}_{(1:t-1)})$

- the next token is unique
- the answer sequence $\mathbf{y}$ is unique

But if we sample according to the probabilities $p(\mathbf{y}_{(t)} \mid \mathbf{y}_{(1:t-1)})$

- there are many possible answer sequences $\mathbf{y}$
- some of them syntactically different but with the same answer
- some with different answers

Thus, it is possible to use compute at test time to generate more than one answer to a prompt.

There are several strategies for using multiple answers to improve Performance

Multiple answer attempts can improve Performance

Suppose we allow the LLM multiple attempts at producing an answer.

Is is likely that one of the candidate answers is the correct one ?

Suppose we sample k answers and have a method for identifying the "best".

How does that affect Performance ?

Define *pass@k* to be true if one of the k candidate answers is correct.

We examine how *pass@k* varies with increasing k.

Coverage vs. number of samples

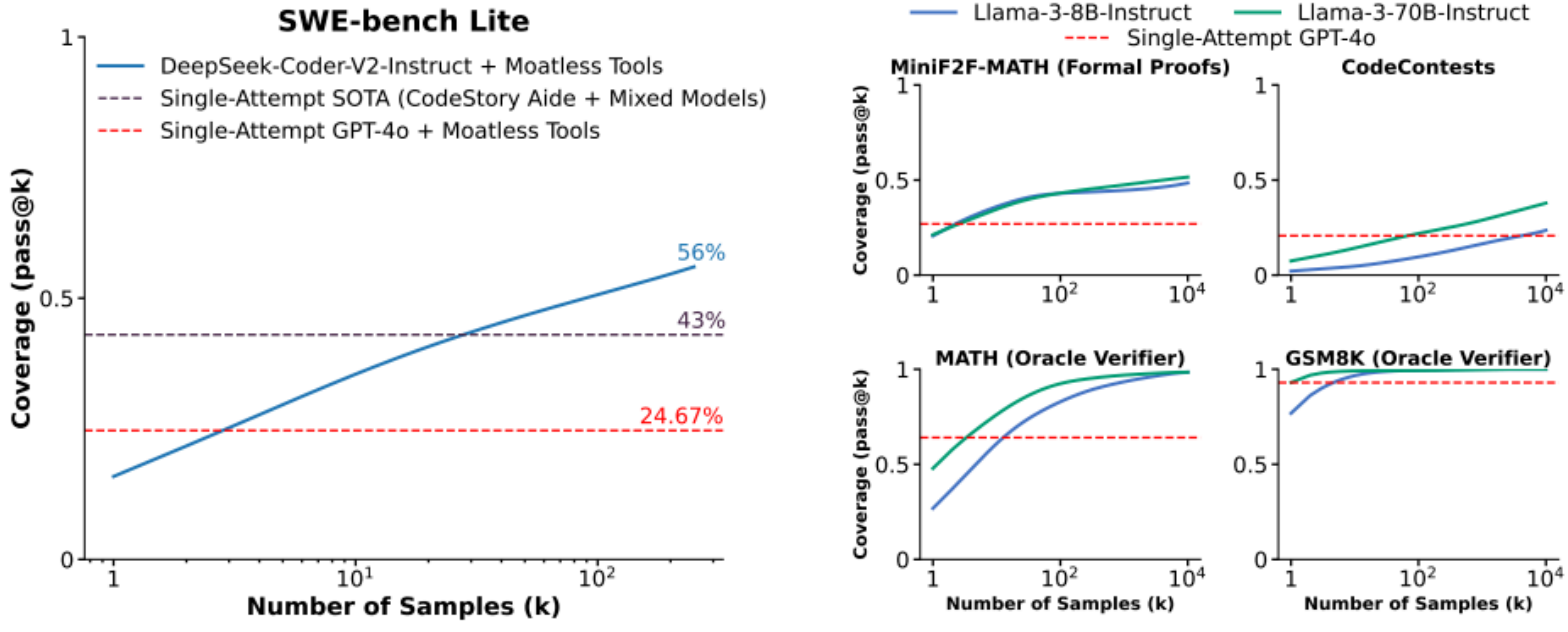


Figure 2: Across five tasks, we find that coverage (the fraction of problems solved by at least one generated sample) increases as we scale the number of samples. Notably, using repeated sampling, we are able to increase the solve rate of an open-source method from 15.9% to 56% on SWE-bench Lite.

Attribution: https://arxiv.org/pdf/2407.21787#page=5

The chart shows that

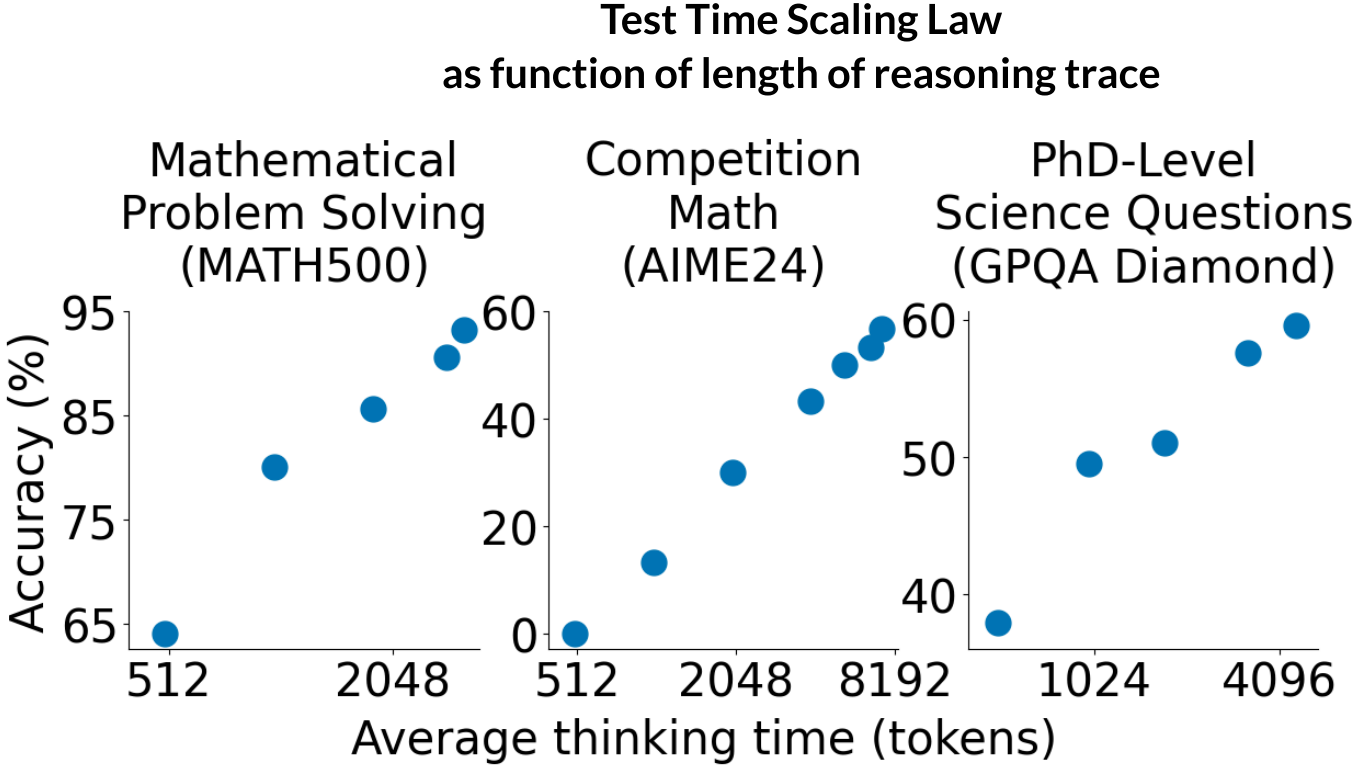
- if we can identify the correct answers from among the k candidates
- Performance increase in a *log-linear* manner
    - a p Percentage Point increase in Performance requires 10^p samples

So test-time compute can be effective, but quite expensive.

Test-time Compute Scaling Law

The simplest description of the Test_time Compute Scaling law is the following chart

- Performance
- versus Compute Budget
    - measured in



Attribution: https://arxiv.org/pdf/2501.19393#page=1 (Note the horizontal scale is logarithmic)

Note that the horizontal axis is $\log_2$ scale.

So the Scaling Law says that

- Performance is log-linear in Compute Budget
- a (constant number of) unit increase in Performance requires *twice* the compute budget

This is very expensive !

Scaling alternatives: Test-time versus Model Size

How does scaling via Test-Time compute

- compare to scaling by increasing the number of parameters ?

The chart below compares the Performance of

- a small model using Test-Time Compute
- a model 14 times larger in parameters that only uses pre-training (no test-time compute)

when both models use the *same compute budget* (FLOPS-matched)

- the small model uses the budget at test-time
- the large model uses the budget only at train-time

Problem instances are assigned "difficulty bins"

- each trace represents one level of difficulty

For each difficulty level

- the traces show the trade-off between Performance and Compute of the test-time compute model
- the horizontal stars of the same color
    - represent the performance of the large (train-compute only) model

When the horizontal stars of one color

- are below the trace of the same color
- spending the compute budget on test-time yields better Performance than spending it at train-time

For the Revisions method (left chart)

- test-time compute yields better Performance
- for the two most difficult classes of examples (purple and red)

To summarize

- there is a log-linear relationship between Performance and Compute
- which is worthwhile for *difficult* problems
    - but for "easier" problems:
        - train-time compute on a larger model
        - is more effective than test-time compute on a model 14 times smaller

Test Time Scaling Law
and
Comparison of using Compute Budget at Train-time (big model) vs Compute-time (small model)
as the difficulty of a problem varies

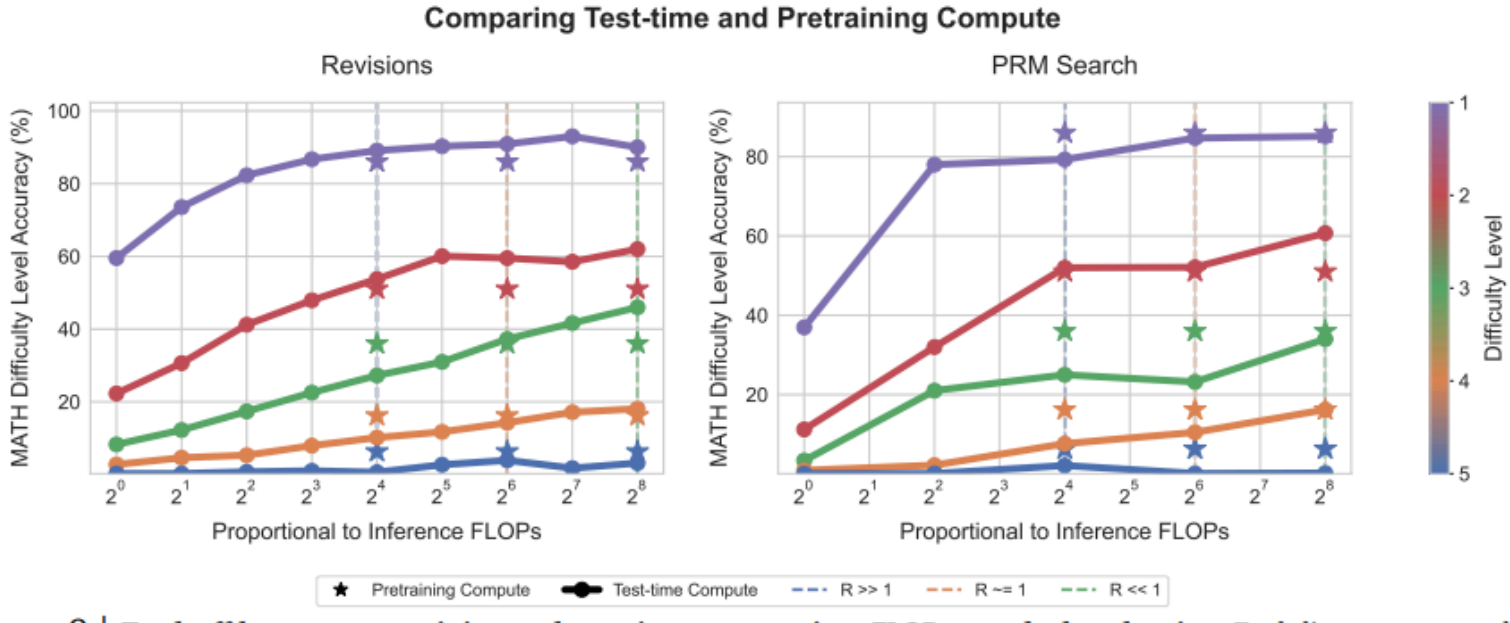


Figure 9 | ***Tradeoff between pretraining and test-time compute in a FLOPs-matched evaluation.*** Each line represents the performance of scaling test-time compute with our compute-optimal policy in each oracle difficulty bin. We plot the results for revisions on the left and search on the right. The stars represent the greedy pass@1 performance of a base model pretrained with ~ 14 times more parameters. We plot test-time compute budget on the x-axis, and place the stars at three different locations along the x-axis, each corresponding to the FLOPs equivalent point of comparison between scaling parameters and scaling test-time compute for three different inference compute loads (e.g. $R = \frac{D_{\text{inference}}}{D_{\text{pretrain}}}$). If the star is below the line, this implies that it is more effective to use test-time compute than to scale model parameters, and if the star is above the line this implies that scaling parameters is more effective. We see that on the easy questions or in settings with a lower inference load (e.g. $R << 1$), test-time compute can generally outperform scaling model parameters. However, on the harder questions or in settings with a higher inference load (e.g. $R >> 1$), pretraining is a more effective way to improve performance.

Attribution: https://arxiv.org/pdf/2408.03314v1#page=15

Strategies for using test-time compute

The above shows the potential for improving Performance by using test-time compute

We examine several common strategies that can be used at test-time to improve Performance.

They fall into two broad classes

- Search
- Reasoning

Search

We use the term "search" to describe

- strategies that explore the space of possible answers

These approaches use an external control process

- to invoke the LLM multiple times
- using logic to control the input to each invocation of the LLM

The model is

- used as a "subroutine" by the external "control" process
- rather than being trained to perform the search directly

Given that a correct answer to our prompt exists: how can we attempt to find it ?

The following chart illustrates some strategies

- to be discussed in subsequent sub-sections

Search Strategies

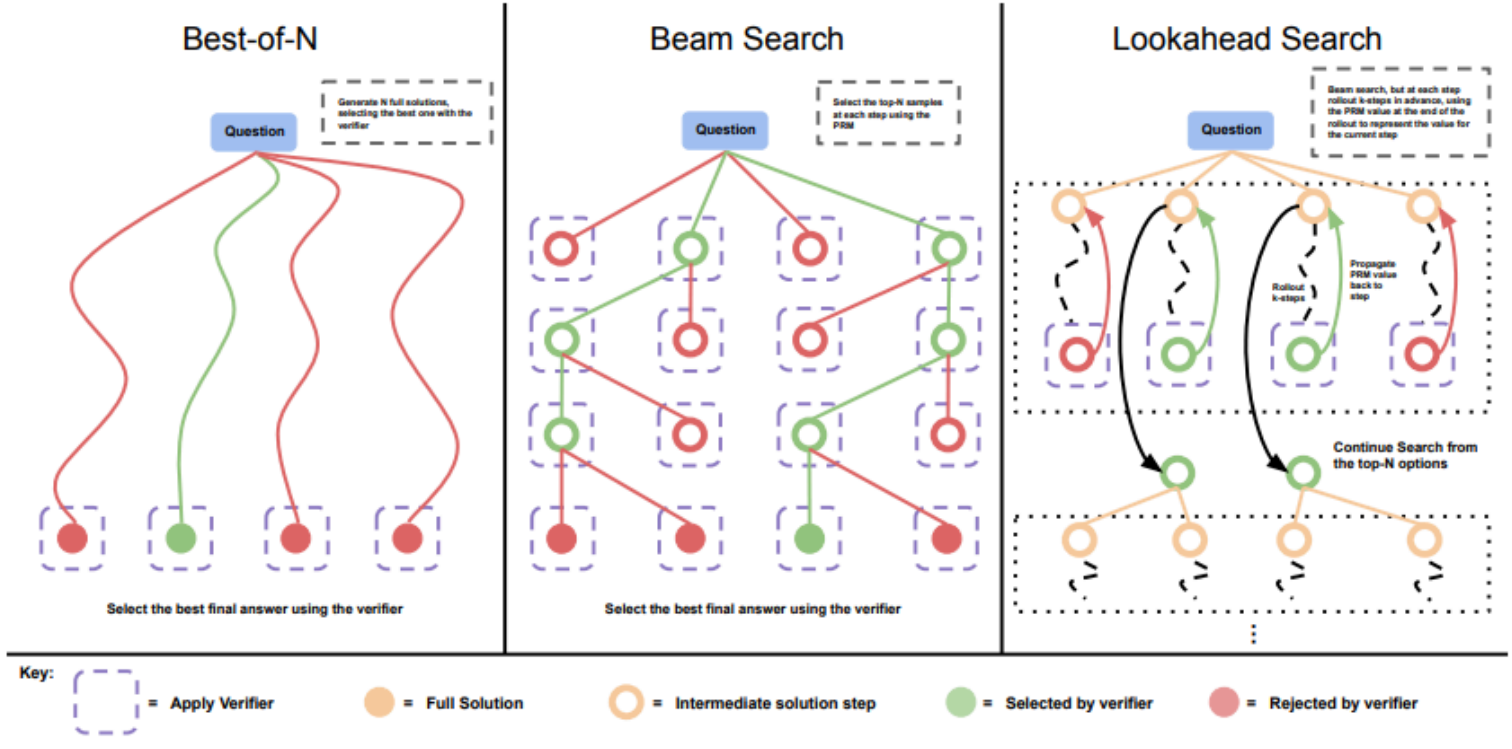


Figure 2 | ***Comparing different PRM search methods.*** **Left:** Best-of-N samples N full answers and then selects the best answer according to the PRM final score. **Center:** Beam search samples N candidates at each step, and selects the top M according to the PRM to continue the search from. **Right:** lookahead-search extends each step in beam-search to utilize a k-step lookahead while assessing which steps to retain and continue the search from. Thus lookahead-search needs more compute.

Attribution: https://arxiv.org/pdf/2408.03314v1#page=8

Some strategies result in multiple candidate solutions.

How do we decide the single final answer ?

That too will be discussed after presenting the strategies for generating the candidates.

Parallel search

Parallel search

- independently samples multiple answers
    - run the non-deterministic LLM multiple times
    - starting from the same initial state (e.g., random seed)

The left-most example in the chart illustrates this approach.

Systematic search

Rather than sampling complete answers independently

- one can imagine a systematic search of the answer space
- generating intermediate (non-final) answers
- deciding which intermediate answers to improve/which to kill

This is a type of parallel approach, where the answers are not independently generated.

This results in a tree-like structure

- where the leaves are the candidate final answers

The center and right-most example in the chart illustrates this approach.

Sequential search via Revision

Revision strategies

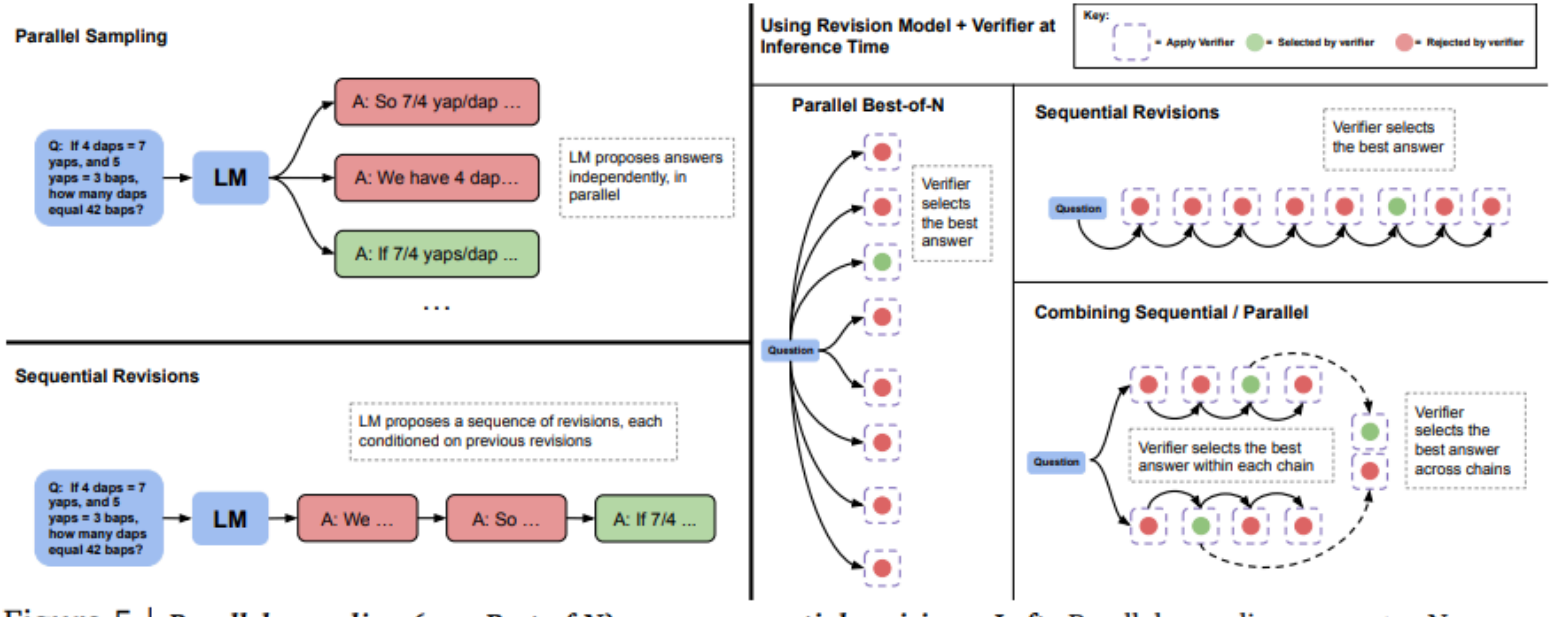


Figure 5 | ***Parallel sampling (e.g., Best-of-N) verses sequential revisions.*** **Left:** Parallel sampling generates N answers independently in parallel, whereas sequential revisions generates each one in sequence conditioned on previous attempts. **Right:** In both the sequential and parallel cases, we can use the verifier to determine the best-of-N answers (e.g. by applying best-of-N weighted). We can also allocate some of our budget to parallel and some to sequential, effectively enabling a combination of the two sampling strategies. In this case, we use the verifier to first select the best answer within each sequential chain and then select the best answer accross chains.

Attribution: https://arxiv.org/pdf/2408.03314v1#page=11

Here we use the LLM multiple times *sequentially* to generate a final answer

- the i^{th} answer is critiqued by the LLM
- and used to create a *revised* successor answer $i + 1$
    - the successor is not necessarily "better", just a result of responding to the critique

Note

- "parallel" revision is just "Best-of-N"

Reducing multiple candidate answers to a single final answer

Some search strategies result in multiple candidate answers.

There are several strategies for reducing to a single, final answer.

Majority vote

Generate N candidates

- select the candidate that occurs most frequently among the N candidates

Using ranking

The following strategies require the ability to rank the candidates

- may be a partial order, rather than a total order

The section on Verifiers expands on this idea.

For now: we show how ranking may be used to reduce the set of candidates.

Best-of-N

Generate N candidates

- select the single "best" candidate as the final answer

Beam Search

This adapts the search to incorporate the ability to rank *partial* answers.

Generate N candidates at each step

- Select the M (the *beam width*) best answers to continue to explore
- Repeat with the M surviving candidates

Verifiers: Identifying the ”best” candidate

Given a collection of candidates, how do we identify the best ?

We create a mechanism called a *Verifier*.

Sampling answers

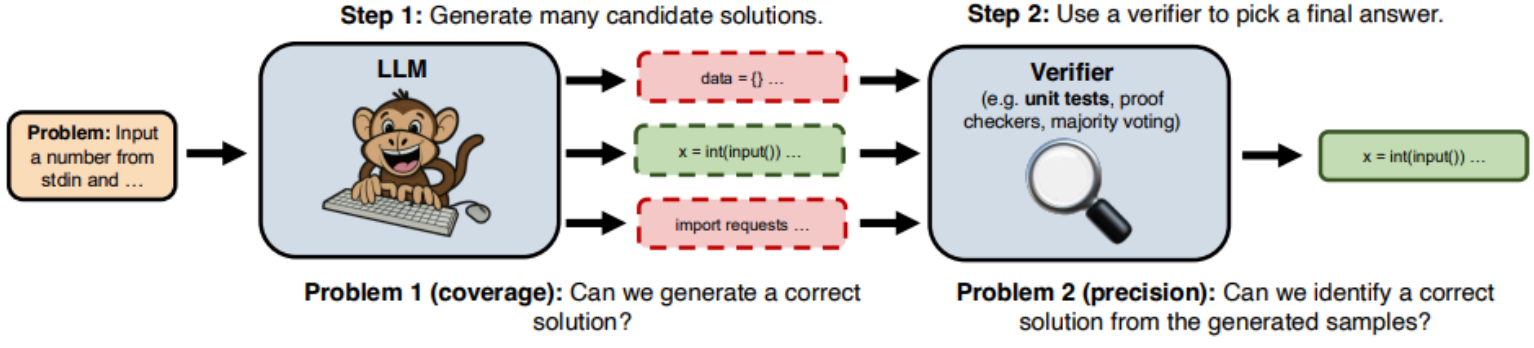


Figure 1: The repeated sampling procedure that we follow in this paper. 1) We generate many independent candidate solutions for a given problem by sampling from an LLM with a positive temperature. 2) We use a domain-specific verifier (ex. unit tests for code) to select a final answer from the generated samples.

Attribution: https://arxiv.org/pdf/2407.21787#page=3

For certain problems

- it is difficult to *efficiently* discover an answer (i.e., not brute-force search)
- but easy to verify whether a candidate is correct

For example

- it is expensive to find Prime numbers
- but easy to verify whether a candidate is Prime

For such problems, it is easy to construct a verifier

- assigns high rank to correct answers

Reward models

For problems where it is difficult/expensive to verify candidate answers, we train a Neural Network to rank candidates

- generating a "reward" for each candidate.

These models are trained using examples created by Human Feedback

- Human prompts an LLM to generate multiple answers to a problem
- Human ranks the answers (really: creates pairs of candidates and identifies the preferred one)
- Neural Network Classifier trained to identify the preferred candidate in the pair

A verifier which is used to verify/rank only *complete* candidate answers

- is called an *Output Reward Model (ORM)*

These can be used to rank the multiple candidates as required by some of the above strategies for producing candidate answers.

More sophisticated reward models

- can be used to *guide* the search
- by evaluating each step in the generation of a single candidate answer.

Such models are called *Process Reward Models (PRM)*

These are commonly used when the answers contain *chains of thought/reasoning*

- a PRM can assign a score to *each reasoning step*
    - can terminate a branch of the search upon discovering an incorrect reasoning step
    - can choose which branches to extend by ranking the quality of the reasoning
        - can be combined with back-tracking

A candidate with a fatally incorrect reasoning step has a low reward.

A candidate with a correct answer has a higher reward

- can combine the per-step rewards into a final reward/score for the candidate
- choose as final answer the candidate with highest reward

Reasoning

The second approach

- trains a model
- to use *strategies*
- that lead to a single final answer

The model produces the answer on its own

- no external control process
- using solution strategies that it has been trained (via examples) to use

In theory

- we could train the model to use the search strategies described above
- given a way to represent the search trace as a sequence
    - LLM's take sequences as input
- we could use a search strategy to generate training examples
- and train the LLM to imitate the strategy

Reasoning models often imitate

- the search strategy: Sequential Search via Revision

This leverages the inherent ability of LLM's

- to use *Chain of Thought* reasoning ("Think step-by-step")
    - generates a sequence of steps (step-by-step)
    - that culminates in a final answer

But a single Chain of Thought (called a *reasoning trace* may not lead to the correct answers.

The model can be trained

- to "reflect on" (i.e., evaluate)
    - intermediate solutions (reasoning traces that have not yet created an answer)
    - potential candidates
- "revise"
    - abandon an intermediate solution
    - change the intermediate solution and extend

This will be the topic of a separate module.

In [2]:
```
print("Done")
```

```
Done
```